

# A Value-Based Review Process for Prioritizing Artifacts

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## ABSTRACT

As a new contribution to Value-based V&V process development, a systematic and multi-criteria process is proposed to quantitatively determine the Value-based V&V artifact priority that reviewers can follow for their reviews. This process enables reviewers to prioritize artifacts to be reviewed in a more cost-effective way based on more sophisticated and comprehensive factors, such as importance, quality risks, dependency and cost of V&V investments. Some qualitative and quantitative evidence is provided from a comparative experiment with 22 real-client e-services projects over two years of a graduate software engineering team-project course. It shows that the value-based artifact prioritization enabled reviewers to better focus on artifacts with high importance and risks, to capture issues with high impact in a timely manner, and to improve the cost-effectiveness of reviews.

## Categories and Subject Descriptors

D.2.9 [Management]: Software process, Software quality assurance (SQA)

## General Terms

Experimentation, Management, Verification, Measurement

## Keywords

Value-based Software Engineering, Verification, Validation, Review

## 1. INTRODUCTION

Most traditional V&V processes usually tend to treat all aspects of V&V activities equally important because of their underlying value-neutral assumption, i.e. such as all artifacts, all sections in an artifact, all concerns or defects discovered being equally important. Consequently, much review effort has been spent on trivial issues like obvious typos and grammatical errors, leaving the relationship between V&V and business decisions unlinked and the potential value contribution of V&V unexploited. However, commercial experience is often that 80% of the business value is covered by 20% of the tests or defects, and that prioritizing by value produces significant payoffs [1-4].

Two key value-based software engineering theorems [5] provide the basis for treating software engineering activities as software investments. They enable software engineers to prioritize their V&V activities to improve their return on investment. As a part of value-based software engineering, the value-based V&V process treats different artifacts, different sections in artifacts and different issues/defects unequally by prioritizing them based on artifacts' importance, quality risks and cost of V&V investments. The Value-based V&V process also identifies concerns, defects, or risks of building the wrong system by understanding the stakeholders' objectives and context while exploring solution approaches. It further enables resolving non-trivial defects early to avoid extensive late rework, and addresses "How much is enough" questions by considering the trade-offs among these multiple factors.

An initial value-based set of peer review guidelines were the subject of a similar experiment [6]. Its process consists of: first, a win-win negotiation among stakeholders [7, 8] defines the priority of each system capability; next, the system capabilities with high priorities were reviewed first; third, at each priority level, the high-criticality sources of risk were reviewed first. The result of the initial experiment found a factor-of-2 improvement in value added per hour of peer review time [6].

The previous process used a set of Value-based checklists which determined the priority of each capability times the criticalities of each section of an artifact to determine the order of review of each capability. Three artifacts, Operational Concept Description (OCD), System and Software Requirement Description (SSRD), System and Software Architecture Description (SSAD) were reviewed in this experiment. However, each artifact had the same priority, leading to a reviewing order that depend only on section criticality values, leaving the prioritization at the artifact level as a potential improvement direction.

This paper presents a systematic and multi-criteria process for quantitatively determining value-based artifact-level review priorities. It summarizes the results of two years of comparative experiments involving 22 teams developing real-client e-services applications in a graduate software engineering real-client project course. Overall, the experiments show that this process improved the cost-effectiveness of reviews.

## 2. STUDY BACKGROUND

At USC, best practices from software engineering industries are introduced to students through a 2-semester graduate software engineering course (Csci577a, b) with real-client projects.

From Fall 2008, the Incremental Commitment Spiral Model (ICSM) [9], a value-based, risk-driven software life cycle process

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model was introduced and tailored as a guideline [10] for this course. It teaches and trains students skills such as understanding and negotiating stakeholder needs, priorities and shared visions; rapid prototyping; evaluating COTS, services options; business and feasibility evidence analysis; and concurrent plans, requirements and solutions development.

In this course, students work in teams and are required to understand and apply the Incremental Commitment Spiral Model for software engineering to real-world projects. In CSCI 577b, student teams develop Initial Operational Capability (IOC) products based on the best results from CSCI 577a. As the guideline for this course, ICSM covers the full system development life cycle based on Exploration, Valuation, Foundations, Development, and Operations phases as shown in Figure 1. The key to synchronizing and stabilizing all of the concurrent product and process definition activities is a set of risk-driven anchor point milestones: the Exploration Commitment Review (ECR), Valuation Commitment Review (VCR), Foundation Commitment Review (FCR), Development Commitment Review (DCR), Rebaselined Development Commitment Review (RDCR), Core Capability Drivethrough (CCD), Transition Readiness Review (TRR), and Operation Commitment Review (OCR). At these milestones, the business, technical, and operational feasibility of the growing package of specifications and plans is evaluated by independent experts. For the course, clients, professors and teaching assistants perform Architecture Review Board (ARB) activities based on [11] to evaluate the package of specifications and plans.

Most off-campus students come from real IT industry with rich experiences. They often take on the roles of Quality Focal Point and Integrated Independent Verification and Validation (IIV&V) to review set of artifacts to find any issues related to completeness, consistency, feasibility, ambiguity, conformance, and risk in order to minimize the issues found at ARB. A series of package review assignments are consecutively given to them after development teams submit their packages during the whole semester. The instructions for each assignment, together with artifact templates in the ICSM Electronic Process Guide (EPG) [10] provide reviewing entry and exit criteria for each package review. Table 1 summarizes the content of the V&V reviews as performed in Fall 2009 and Fall 2010, and Table 2 gives the definitions of the ICSM and all other acronyms used in this paper. Details of the review process are in Section 4.

**Table 1.V&V assignments for Fall2009/2010**

| V&Ver Assignment                                   | Review Package                                                         | 2009 V&V Method | 2010 V&V Method |
|----------------------------------------------------|------------------------------------------------------------------------|-----------------|-----------------|
| Learn to Use Bugzilla System for Your Project Team |                                                                        |                 |                 |
| Eval of VC Package                                 | OCD,FED, LCP                                                           | FV&V            | FV&V            |
| Eval of Initial Prototype                          | PRO                                                                    | FV&V            | FV&V            |
| Eval of Core Package                               | OCD,PRO,SSRD**,SSAD,LCP,FED, SID                                       | FV&V            | VbV&V           |
| Eval of Draft FC Package                           | OCD,PRO,SSRD**,SSAD,LCP,FED, SID                                       | FV&V            | VbV&V           |
| Eval of FC/DC Package                              | OCD,PRO,SSRD**,SSAD,LCP,FED, SID, QMP, ATPC^, IP^                      | FV&V            | VbV&V           |
| Eval of Draft DC/TRR Package                       | OCD,PRO,SSRD**,SSAD,LCP,FED, SID, QMP, ATPC^, IP^, TP^                 | VbV&V           | VbV&V           |
| Eval of DC/TRR Package                             | OCD,PRO,SSRD**,SSAD,LCP,FED, SID, QMP, ATPC, IP, TP, IAR^,UM^,TM^,TPR^ | VbV&V           | VbV&V           |
| **: not required by NDI/NCS team;                  |                                                                        |                 |                 |
| ^: only required by one-semester team;             |                                                                        |                 |                 |

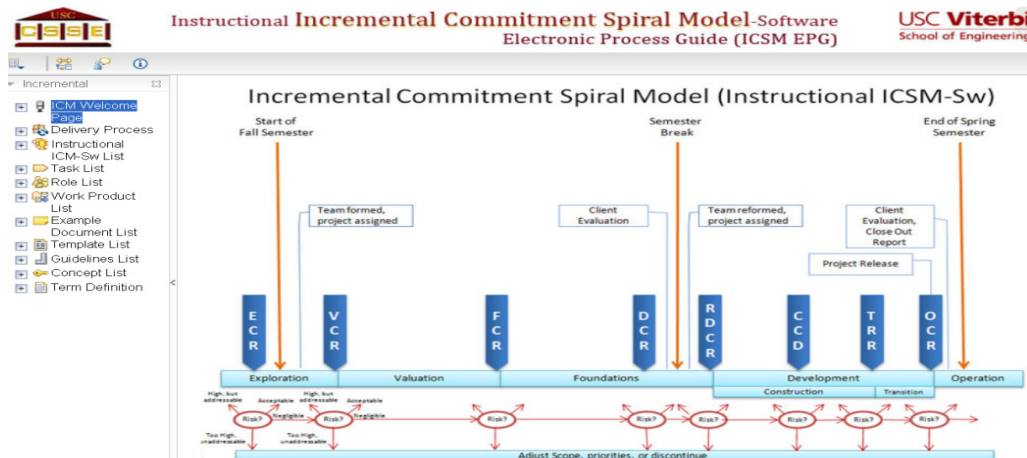
**Table 2. Acronyms**

•**ICSM phases**: **VC**: Valuation Commitment, **FC**: Foundation Commitment, **DC**: Development Commitment, **TRR**: Transition Readiness Review, **RDC**: Rebaselined Development Commitment, **IOC**: Initial Operational Capability, **TS**: Transition & Support

•**Artifacts** developed and reviewed for this course:

**OCD**: Operational Concept Description, **SSRD**: System and Software Requirements Description, **SSAD**: System and Software Architecture Description, **LCP**: Life Cycle Plan, **FED**: Feasibility Evidence Description, **SID**: Supporting Information Document, **QMP**: Quality Management Plan, **IP**: Iteration Plan, **IAR**: Iteration Assessment Report, **TP**: Transition Plan, **TPC**: Test Plan and Cases, **TPR**: Test Procedures and Result, **UM**: User Manual, **SP**: Support Plan, **TM**: Training Materials

•**Others**: **FV&V**: Formal Verification & Validation, **VbV&V**: Value-based Verification & Validation, **Eval**: Evaluation, **ARB**: Architecture Review Board



**Figure 1. ICSM framework tailored for csci577 [10]**

### 3. COMPARATIVE V&V PROCESSES

#### 3.1 Value-neutral Formal V&V Process

As displayed in Table 3, for the first few evaluation package review assignments, students are taught and trained with a formal V&V process (FV&V), a variant of Fagan inspection [12] practice. The steps they followed are:

**Table 3. Instructions for Formal V&V process**

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Step 1: Create Exit Criteria:</b> From the original team assignment’s description and the related ICSM EPG completion criteria, generate a set of exit criteria that identify what needs to be present and the standard for acceptance of each document.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Step 2: Review and Report Concerns:</b> Based upon the exit criteria, read (review) the documents and report concerns and issues into the Bugzilla [13] system.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Step 3: Generate Evaluation Report</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <ul style="list-style-type: none"> <li>•Management Overview - List any features of the solution described in this artifact that are particularly good, of which a non-technical client should be aware of.</li> <li>•Technical Details - List any features of the solution described in this artifact that you feel are particularly good, and which a technical reviewer should be aware of.</li> <li>•Major Errors &amp; Omissions - List top 3 errors or omissions in the solution described in this artifact that a non-technical client would care about. The description of an error (or omission) should be understandable to a non-technical client, and should explain why the error is worth the client’s attention.</li> <li>•Critical Concerns - List top 3 concerns with the solution described in this artifact that a non-technical client would care about. The description of the concern should be understandable to a non-technical client, and should explain why the client should be aware of it. You should also suggest step(s) to take that would reduce or eliminate your concern.</li> </ul> |

#### 3.2 Value-based V&V Process

A Value-based V&V Process (VbV&V) is then introduced to the course for the later review package assignments. As a part of value-based software engineering, the value-based V&V process treats different artifacts, different sections in artifacts and different issues/defects as unequally important. Besides the importance to project’s success, several other factors should be considered for this prioritization. For example, it is reasonable to delay reviewing an artifact with high importance but fewest estimated defects and focus earlier on the fault-prone ones with relatively lower importance. Also the dependencies among artifacts make first reviewing some less-important but necessary to more-important ones. In addition, a higher priority may be given to reviewing the ones requiring less reviewing effort when the reviewing time is very limited.

Besides, the concurrent and incremental artifact development makes the artifact/system capability reviewing prioritization not a static and once for all process. The priority initially negotiated by stakeholders might change with large uncertainty in later developing phases, which makes refining and updating the reviewing priority before each review necessary and valuable.

Considering those factors that would influence the reviewing priority for artifacts, a systematic and multi-criteria process is proposed for Value-based V&V artifact prioritization as shown in Table 4. Before each package review, the prioritization should be generated or updated based on the current project’s context and situation.

**Table 4. Instructions for Value-based V&V process**

| <b>Step 1: Value-based V&amp;V Artifacts Prioritization</b>                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Priority Factor                                                                                    | Rating Guideline                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Importance</b><br>5: most important<br>3: normal<br>1: least important                          | <ul style="list-style-type: none"> <li>•Without this document, the project can’t move forward or could even fail; it should be rated with high importance</li> <li>•Some documents serve a supporting function. Without them, the project still could move on; this kind of document should be rated with lower importance</li> </ul>                                                                                                                                                                                  |
| <b>Quality Risk</b><br>5: highly risky<br>3: normal<br>1: least risky                              | <ul style="list-style-type: none"> <li>•Based on previous reviews, the documents with intensive defects might be still fault-prone, so this indicates a high quality risk</li> <li>•Personnel factors, e.g. the author of this documents is not proficient or motivated enough; this indicates a high quality risk</li> <li>•A more complex document might have a high quality risk</li> <li>•A new document or an old document with a large portion of newly added sections might have a high quality risk</li> </ul> |
| <b>Dependency</b><br>5: highly dependent<br>3: normal<br>1: not dependent                          | <ul style="list-style-type: none"> <li>•Sometimes some lower-priority artifacts are required to be reviewed at least for reference before reviewing a higher-priority one. For example, in order to review SSAD or TPC, SSRD is required for reference.</li> <li>•Basically, the more documents this document depends on, the higher the Dependency rating is, and the lower the reviewing priority will be.</li> </ul>                                                                                                |
| <b>Review Cost</b><br>5: need intensive effort<br>3: need moderate effort<br>1: need little effort | <ul style="list-style-type: none"> <li>•A new document or an old document with a large portion of newly added sections usually takes more time to review and vice versa</li> <li>•A more complex document usually takes more time to review and vice versa</li> </ul>                                                                                                                                                                                                                                                  |
| <b>Determine Weights</b>                                                                           | <ul style="list-style-type: none"> <li>•Weights for each factor (Importance, Quality Risk, Review Cost, and Dependency) could be set according to the project context. Default values are 1.0 for each factor</li> </ul>                                                                                                                                                                                                                                                                                               |

### Priority Calculation

#### Priority

$$= \frac{(W_I * Importance) * (W_{QR} * QualityRisk)}{(W_D * Dependency) * (W_{RC} * ReviewingCost)}$$

E.g: for a document, Importance=5, Quality Risk=3, Review Cost=2, Dependency = 1, default weights are used=>

$$Priority = (5*3)/(2*1)=7.5$$

•A spreadsheet [14] helps to calculate the priority automatically, 5-level ratings for each factor are VH, H, M, L VL with values from 5 to 1, intermediate values 2, 4 are also allowed.

#### Step 2: Review artifacts based on prioritization and report defects/issues

- The one with higher priority value should be reviewed first
- For each document's review, review the core part of the document first. Report issues into the Bugzilla [13]

#### Step 3: List top 10 defects/ issues

- List top 10 highest-risk defects or issues based on issues' priority and severity

A real example of artifacts prioritization in CoreFCP evaluation by a 2010-team [15] is displayed in Table 16 at the end of the paper. The default weight of 1.0 for each factor is used. Based on the priority calculated, reviewing order follows SSRD, OCD, PRO, SSAD, LCP, FED, SID. SSRD has the highest reviewing priority with the rationales provided: SSRD contains the requirements of the system, without this document, the project can't move forward or could even fail (Very-High Importance). This is a complex document, and needs to be consistent with win conditions negotiation, which might not be complete at this point, also, a lot of rework was required based on comments from TA (Very-High Quality Risk). SSRD depends on few other artifacts (Low Dependency). This is an old document, but it is complex with a lot of rework (Very-High Review Cost).

An example of Top 10 issues made by this team for CoreFCP evaluation is displayed in Table 5. These Top 10 issues are communicated in a timely manner with artifact authors to attract enough emphasis. Interesting finding is the relations between the artifact priority sequence and the top 10 issues sequence: the issues with higher impact usually exist in the artifacts with high priority, showing that the artifact prioritization enables reviewers to focus on issues with high impact at least in this context. However, it also helps avoid the potential problem of neglecting high-impact issues in lower-priority artifacts, as in Issues 8 and 10.

**Table 5. An example of Top 10 Issues**

|    |                  | Summary                                                                                        | Rationale                                                                                                                                                                                                                                                         |
|----|------------------|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | S<br>S<br>R<br>D | Missing important requirements.                                                                | A lot of important requirements are missing. Without these requirements, the system will not succeed.                                                                                                                                                             |
| 2  | S<br>S<br>R<br>D | Requirement supporting information too generic.                                                | The output, destination, precondition, and post condition should be defined better. These description will allows the development team and the client better understand the requirements. This is important for system success.                                   |
| 3  | S<br>S<br>A<br>D | Wrong cardinality in the system context diagram.                                               | The cardinality of this diagram needs to be accurate since this describes the top level of the system context. This is important for system success.                                                                                                              |
| 4  | O<br>C<br>D      | The client and client advisor stakeholders should be concentrating on the deployment benefits. | It is important for that this benefits chain diagram accurately shows the benefits of the system during deployment in order for the client to show to potential investor to gather fund to support the continuation of system development.                        |
| 5  | O<br>C<br>D      | The system boundary and environment missing support infrastructure.                            | It is important for the System boundary and environment diagram to capture all necessary support infrastructure in order for the team to consider all risks and requirements related the system support infrastructure.                                           |
| 6  | F<br>E<br>D      | Missing use case references in the FED.                                                        | Capability feasibility table proves the feasibility of all system capabilities to date. Reference to the use case is important for the important stakeholders to understand the capabilities and their feasibility.                                               |
| 7  | F<br>E<br>D      | Incorrect mitigation plan.                                                                     | Mitigation plans for project risks are important to overcome the risks. This is important for system success.                                                                                                                                                     |
| 8  | L<br>C<br>P      | Missing skills and roles                                                                       | The LCP did not identify the skill required and roles for next semester. This information is important for the success of the project because the team next semester can use these information and recruit new team members meeting the identified needed skills. |
| 9  | F<br>E<br>D      | CR# in FED doesn't match with CR# in SSRD                                                      | The CR numbers need to match in both FED and SSRD for correct requirement references.                                                                                                                                                                             |
| 10 | L<br>C<br>P      | COCOMO drivers rework                                                                          | COCOMO driver values need to be accurate to have a better estimate for the client.                                                                                                                                                                                |

## 4. EXPERIMENT DESIGN

There were 14 development teams in Fall 2009 and 8 teams in Fall 2010 [16, 17] with 7-8 persons per team (6 on-campus and 1-2 off-campus students). Off-campus students often take on the roles of Quality Focal Point and Integrated Independent Verification and Validation (IIV&V) to review packages before ARB to minimize the concerns that found at ARB. 7 package review assignments [18, 19] as shown in Table 7 were given to them consecutively after development teams submitted their packages during the fall semester.

For both years, The Formal V&V (FV&V) process was introduced for the first few package reviews, and then the Value-based V&V (VbV&V) process was introduced to the course for the later package reviews. Later a comparative survey briefly shown in Table 6 is assigned to reviewers to compare the Formal V&V process and Value-based V&V process, and allow students to freely choose the process that they think is more effective with rationales. They were also be notified that their process choices were irrelevant with their assignments grades. This survey provides qualitative and real-life experiences and evidence from the real users that adopted Value-based V&V and the outcome of the survey will be introduced in section 5.1.

The difference between the two years is that in Fall 2009, the Value-based V&V process was introduced from the Evaluation of DraftDC DC/TRR Package, but in Fall 2010, the introduction of Value-based V&V process was brought forward to Evaluation of Core FC Package as displayed in Table 1. So under this situation, we have the chance to compare the review effectiveness measures such as Average of Impact per Issue, Review Cost Effectiveness of Issue as shown in Table 10 in section 5.2 for Core FC Package (CoreFCP), Draft FC Package (DraftFCP), and FC/DC Package (FC/DCP) with independent 14 2009 teams using the value-neutral Formal V&V method and 8 2010 teams using the Value-based V&V method.

**Table 6. Instructions for process selection**

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**Choose process and give rationales:** In your last package review, a Value-based IIVV Process is used. For this package review, you are allowed to freely choose either Value-based V&V Process or Formal V&V Process, and give rationales.

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One of the principles underlying the ICSM is concurrent definition and spiral development. In this course, most of the documents are developed and refined concurrently and incrementally during the whole life cycle. Table 7 shows the documents and sections to be reviewed in CoreFCP, DraftFCP and FC/DCP. The package for evaluation is growing as the project goes on. Different best-fit development processes are: Architected Agile (AA, agile methods scaled with early investment in a largely change-prescient architecture and user/developer/customer team building), Net-Centric Services (NCS, mainly includes Web services, Web application, and online application), Non Developmental Item (NDI, mainly includes COTS, Open Source Software, Reuse library and Customer-furnished package). The estimated duration (1 or 2 semesters) would also influence the documents and sections to be reviewed.

**Table 7. Documents and sections to be reviewed**

| Doc/Sec     | CoreFCP                                             | DraftFCP             | FC/DCP               |                      |
|-------------|-----------------------------------------------------|----------------------|----------------------|----------------------|
|             | 1&2 sem                                             | 1&2 sem              | 2 sem                | 1 sem                |
| <b>OCD</b>  | 100%                                                | 100%                 | 100%                 | 100%                 |
| <b>FED</b>  | AA(Section 1,5)<br>NDI(Section 1,3,4.1,4.2.1,4.2.2) | Section 1-5          | Section 1-5          | 100%                 |
| <b>LCP</b>  | Section 1, 3.3                                      | 100%                 | 100%                 | 100%                 |
| <b>SSRD</b> | AA(100%)<br>NDI(N/A)                                | AA(100%)<br>NDI(N/A) | AA(100%)<br>NDI(N/A) | AA(100%)<br>NDI(N/A) |
| <b>SSAD</b> | Section 1, 2.1.1-2.1.3                              | Section 1, 2         | Section 1, 2         | 100%                 |
| <b>PRO</b>  | Most critical/im portant use cases                  | 100%                 | 100%                 | 100%                 |
| <b>SID</b>  | 100%                                                | 100%                 | 100%                 | 100%                 |
| <b>QMP</b>  | N/A                                                 | N/A                  | Section 1,2          | 100%                 |
| <b>ATPC</b> | N/A                                                 | N/A                  | N/A                  | 100%                 |
| <b>IP</b>   | N/A                                                 | N/A                  | N/A                  | 100%                 |

## 5. EXPERIMENT RESULTS

In order to comprehensively compare the effectiveness of value-neutral and value-based V&V processes, both qualitative evidence and quantitative comparison are displayed in this section.

### 5.1 Qualitative Evidence

In the survey after introducing the Value-based V&V artifact prioritization process, all 14 2009-teams and 8 2010-teams chose the Value-based reviewing process. Various advantages are identified by reviewers, such as: more streamlined, efficient, not a waste of time, more focused on most important documents with high quality risks, more focused on non-trivial defects and issues, an organized and systematic way to review documents in an integrated way, not treating documents independently. Some example responses are as below:

*“The value-based V&V approach holds a great appeal – a more intensive and focused V&V process. Since items are prioritized and rated as to importance and likelihood of having errors. This is meant for you to allocate your time according to how likely errors (and how much damage could be done) will occur in an artifact. By choosing to review those areas that have changed or are directly impacted by changes in the other documents I believe I can give spend more quality time in reviewing the changes and give greater emphasis on the changes and impacts.”*

*“Top 10 issue list gives a centralized location for showing the issues as opposed to spread across several documents. Additionally, by prioritizing the significance of each issue, it gives document authors a better picture of which issues they should spend more time on resolving and let them know which ones are more important to resolve. Previously, they would have just tackled the issues in any particular order, and may not have spent*

the necessary time or detail to ensure proper resolution. Focusing on a top 10 list helps me to look at the bigger picture instead of worrying about as many minor problems, which will result in documents that will have fewer big problems.”

“For the review of the Draft FC Package, the Value-based IIV&V Process will be used. This review process was selected because of the time constraint of this review. There is only one weekend to review all seven Draft FC Package documents. The Value-based review will allow me to prioritize the documents based on importance, quality risk, dependencies, and reviewing cost. The documents will be reviewed based on its identified priority. This allows documents more critical to the success of the project to be reviewed first and given more time to. ”

These responses and the unanimous choice of using the Value-based process show that the performers considered the Value-based V&V process to be superior to the formal V&V process for achieving their project objectives.

## 5.2 Quantitative Comparison

### 5.2.1 Data Collection

The two-year experiment issue data for the evaluation of CoreFCP, DraftFCP and FC/DCP from total 22 teams is collected and extracted from the Bugzilla MySQL database.

### 5.2.2 Data Transformation

The generic term “Issue” covers both “Concerns” and “Problems”. If the IV&Vers find any issue, they report it as a “Concern” in Bugzilla and assign it to the relevant artifact author. The author determines whether it needs fixing by choosing an option for “Resolution” as displayed in Table 8. Whether an issue is a problem or not is easy to be determined by querying the “Resolution” of the issue. “Fixed” and “Won’t Fix” mean the issue is a problem and the other two options mean that it is not.

**Table 8. Resolution options in Bugzilla**

| Resolution Options | Instructions in Bugzilla                                                                                                                                                            |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fixed              | If the issue is a problem, after you fix the problem in the artifact, then choose “Fixed”                                                                                           |
| Won’t Fix          | If the issue is a problem, but won’t be fixed for this time, then choose “Won’t Fix” and must provide the clear reason in “Additional Comments” why it can’t be fixed for this time |
| Invalid            | If the issue is not a problem then choose “Invalid” and must provide a clear reason in “Additional Comments”                                                                        |
| WorksForMe         | If the issue really works fine, then choose “WorksForMe” and let the IVVer review this again                                                                                        |

The main goal of Value-based review is to increase the cost effectiveness of the review [5]. In other words, it is to improve the Return on Investment (ROI) of reviewing activities. The value of a review could be determined by the number of issues found by this review and how severe and urgent these issues are. To quantify the value (effectiveness) of a review, Review Effectiveness and Cost Effectiveness are defined as below:

$$\text{Eq.1: Review Effectiveness} = \sum_{\text{issues}} \text{Issue Impact}$$

$$= \sum_{\text{issues}} \text{Issue Severity} * \text{Issue Priority}$$

$$\text{Eq. 2: Review Cost Effectiveness (ROI)} = \frac{\text{Review Effectiveness}}{\text{Effort}}$$

The Severity and Priority are rated respectively from High to Low with the value from 3 to 1. The Impact of an issue is the product of its Severity and Priority. The impact of an issue with high severity and high priority is 9. Obviously, the impact of an issue is an element in the set {1, 2, 3, 4, 6, and 9}. The relevant rating levels in Bugzilla mapped to the three level rating is displayed in Table 9.

**Table 9. Issue Severity & Priority rate mapping**

|          | Rating for Measurement | Rating in Bugzilla                       | Value |
|----------|------------------------|------------------------------------------|-------|
| Severity | High                   | Blocker, Critical, Major                 | 3     |
|          | Medium                 | Normal                                   | 2     |
|          | Low                    | Minor, Trivial, Enhancement              | 1     |
| Priority | High                   | Resolve Immediately                      | 3     |
|          | Medium                 | Normal Queue                             | 2     |
|          | Low                    | Not Urgent, Low Priority, Resolved Later | 1     |

### 5.2.3 Comparative Analysis

In order to quantitatively explore whether the Value-based Review is superior to the formal value-neutral one or not, various measures are adopted in Table 10 to compare their performance.

**Table 10. Review effectiveness measures**

| Measures                              | Details                                                                              |
|---------------------------------------|--------------------------------------------------------------------------------------|
| Number of Concerns                    | The number of concerns found by reviewers                                            |
| Number of Problems                    | The number of problems found by reviewers                                            |
| Number of Concerns per reviewing hour | The number of concerns found by reviewers per reviewing hour                         |
| Number of Problems per reviewing hour | The number of problems found by reviewers per reviewing hour                         |
| Review Effort                         | Effort spent on all activities as shown in Table 3 and Table 4 in the package review |

|                                        |                                                            |
|----------------------------------------|------------------------------------------------------------|
| Review Effectiveness of total Concerns | As defined as Eq.1 but for concerns                        |
| Review Effectiveness of total Problems | As defined as Eq.1 but for problems                        |
| Average of Impact per Concern          | Review Effectiveness of total Concerns/ Number of Concerns |
| Average of Impact per Problem          | Review Effectiveness of total Problems/ Number of Problems |
| Review Cost Effectiveness of Concerns  | As defined as Eq.2 but for concerns                        |
| Review Cost Effectiveness of Problems  | As defined as Eq.2 but for problems                        |

For each measure, Mean and Standard Deviation values are compared between the two-year teams. The improvement percentage of Value-based review team from Value-neutral team in terms of %2010 Team higher is calculated by the Mean value difference divided by 2009 team's Mean value.

To determine whether the improvement based on this measure is statistically significant or not, the F-test and T-test are used. The F-test determines whether two samples have different variances. If the significance (p-value) for F-test is 0.05 or below, the two samples have different variances. This will determine which type of T-test will be used to determine whether the two samples have the same mean. Two types of T-test are: Two-sample equal variance (homoscedastic), and Two-sample unequal variance (heteroscedastic). If the significance (p-value) for T-test is 0.05 or below, the two samples have different means.

An example of comparative analysis between the two-year teams based on the measure Review Cost Effectiveness of Concerns is shown in Table 11. The Mean, Standard Deviation and %2010 Team higher are calculated. This shows that 2010's value-based review teams had a 75.04% higher Review Cost Effectiveness of Concerns than 2009's value-neutral teams. The p-value for F-test 0.0060 leads to choose "Two-sample unequal variance" type T-test. The p-value for T-test 0.0218 is strong evidence (well below 0.05) that the 75.04% improvement has statistical significance.

**Table 11. Comparative analysis example 1**

| Review Cost Effectiveness of Concerns |       |         |      |
|---------------------------------------|-------|---------|------|
| 2010-1                                | 15.64 | 2009-1  | 3.40 |
| 2010-2                                | 13.50 | 2009-2  | 5.15 |
| 2010-3                                | 5.36  | 2009-3  | 7.03 |
| 2010-4                                | 4.98  | 2009-4  | 5.72 |
| 2010-5                                | 4.21  | 2009-5  | 3.73 |
| 2010-6                                | 12.50 | 2009-6  | 3.76 |
| 2010-7                                | 11.90 | 2009-7  | 6.89 |
| 2010-8                                | 6.31  | 2009-8  | 7.38 |
|                                       |       | 2009-9  | 8.76 |
|                                       |       | 2009-10 | 4.28 |
|                                       |       | 2009-11 | 6.55 |
|                                       |       | 2009-12 | 5.96 |
|                                       |       | 2009-13 | 2.92 |
|                                       |       | 2009-14 | 2.85 |

|                                                |      |       |      |
|------------------------------------------------|------|-------|------|
| Mean                                           | 9.30 | Mean  | 5.31 |
| Stdev                                          | 4.53 | Stdev | 1.86 |
| % 2010 Team higher:<br>(9.30-5.31)/5.31=75.04% |      |       |      |
| F-test (p-value): 0.0060                       |      |       |      |
| T-test (p-value): 0.0218                       |      |       |      |

Another example is the comparative analysis based on the measure Average of Impact per Concern as shown in Table 12. This shows that 2010's value-based review teams had an 11.14% higher Average of Impact per Concern than 2009's value-neutral teams. The p-value for F-test 0.5957 leads to choose "Two-sample equal variance" type T-test. The p-value for T-test 0.0230 is strong evidence that the 11.14% improvement also has statistical significance. This rejects the hypothesis that the Value-based review method would not enable reviewers to find more high impact concerns.

**Table 12. Comparative analysis example 2**

| Average of Impact per Concern                  |      |         |      |
|------------------------------------------------|------|---------|------|
| 2010-1                                         | 5.60 | 2009-1  | 4.17 |
| 2010-2                                         | 4.40 | 2009-2  | 4.13 |
| 2010-3                                         | 4.40 | 2009-3  | 3.27 |
| 2010-4                                         | 4.45 | 2009-4  | 4.01 |
| 2010-5                                         | 3.88 | 2009-5  | 4.72 |
| 2010-6                                         | 4.14 | 2009-6  | 3.21 |
| 2010-7                                         | 4.12 | 2009-7  | 4.72 |
| 2010-8                                         | 4.32 | 2009-8  | 3.55 |
|                                                |      | 2009-9  | 4.02 |
|                                                |      | 2009-10 | 3.75 |
|                                                |      | 2009-11 | 4.09 |
|                                                |      | 2009-12 | 3.90 |
|                                                |      | 2009-13 | 3.91 |
|                                                |      | 2009-14 | 4.15 |
| Mean                                           | 4.42 | Mean    | 3.97 |
| Stdev                                          | 0.52 | Stdev   | 0.44 |
| % 2010 Team higher:<br>(4.42-3.97)/3.97=11.14% |      |         |      |
| F-test (p-value): 0.5957                       |      |         |      |
| T-test (p-value): 0.0230                       |      |         |      |

Another example is the comparative analysis based on the measure Number of Concerns as shown in Table 13. Although 2010's value-based review teams increase 33.69% Number of Concerns than 2009's value-neutral teams, the p-value for T-test 0.1093 shows that this increase does not have statistical significance, i.e., that the difference in the Number of Concerns between the two years is within the range of uncertainty.

**Table 13. Comparative analysis example 3**

| Number of Concerns |     |         |     |
|--------------------|-----|---------|-----|
| 2010-1             | 141 | 2009-1  | 58  |
| 2010-2             | 198 | 2009-2  | 45  |
| 2010-3             | 53  | 2009-3  | 102 |
| 2010-4             | 33  | 2009-4  | 87  |
| 2010-5             | 60  | 2009-5  | 32  |
| 2010-6             | 116 | 2009-6  | 58  |
| 2010-7             | 98  | 2009-7  | 103 |
| 2010-8             | 94  | 2009-8  | 119 |
|                    |     | 2009-9  | 157 |
|                    |     | 2009-10 | 61  |
|                    |     | 2009-11 | 108 |
|                    |     | 2009-12 | 41  |

|       |       |         |       |
|-------|-------|---------|-------|
|       |       | 2009-13 | 34    |
|       |       | 2009-14 | 33    |
| Mean  | 99.13 | Mean    | 74.14 |
| Stdev | 53.28 | Stdev   | 38.75 |

%2010 Team higher:

$$(99.13-74.14)/74.14=33.69\%$$

F-test (p-value): 0.3049

T-test (p-value): 0.1093

The comparative analysis results for all measures in Table 10 are displayed in Table 14 and Table 15. If the improvement based on a measure is statistically significant (p-value for T-test is 0.05 or below), it is highlighted with shadow in Table 15. From these results, it is clear that although the differences based on Number of Concerns/Problems and Review Effort are not statistically significant, the differences based on measures Number of Concerns/ Problems per reviewing hour, Review Effectiveness of total Concerns/Problems, Average of Impact per Concern/Problem, and Review Cost Effectiveness of Concerns/Problems are all significant. These comparative analysis results show that the Value-based review method to prioritize artifacts can improve the cost effectiveness of reviewing activities, and can enable reviewers to be more focused on artifacts with high importance and risks, and capture concerns/problems with high impact.

**Table 14. Comparative results**

|                                        | 2010 Team |        | 2009 Team |        |
|----------------------------------------|-----------|--------|-----------|--------|
|                                        | Mean      | Stdev  | Mean      | Stdev  |
| Number of Concerns                     | 99.13     | 53.28  | 74.14     | 38.75  |
| Number of Problems                     | 93.38     | 52.96  | 68.79     | 35.35  |
| Number of Concerns per reviewing hour  | 2.08      | 0.93   | 1.36      | 0.52   |
| Number of Problems per reviewing hour  | 1.96      | 0.92   | 1.26      | 0.48   |
| Review Effort                          | 47.51     | 13.37  | 53.35     | 13.97  |
| Review Effectiveness of total Concerns | 445.63    | 263.08 | 292.00    | 155.05 |
| Review Effectiveness of total Problems | 416.25    | 254.15 | 272.07    | 141.78 |
| Average of Impact per Concern          | 4.42      | 0.52   | 3.97      | 0.44   |
| Average of Impact per Problem          | 4.37      | 0.52   | 3.99      | 0.45   |
| Review Cost Effectiveness of Concerns  | 9.30      | 4.53   | 5.31      | 1.86   |
| Review Cost Effectiveness of Problems  | 8.69      | 4.32   | 4.97      | 1.73   |

**Table 15. Comparative results (continued)**

|                                        | % 2010 Team higher | F-test (p-value) | T-test (p-value) |
|----------------------------------------|--------------------|------------------|------------------|
| Number of Concerns                     | 33.69%             | 0.3049           | 0.1093           |
| Number of Problems                     | 35.75%             | 0.1976           | 0.1026           |
| Number of Concerns per reviewing hour  | 53.17%             | 0.0636           | 0.0142           |
| Number of Problems per reviewing hour  | 55.16%             | 0.0393           | 0.0382           |
| Review Effort                          | -10.94%            | 0.9509           | 0.1752           |
| Review Effectiveness of total Concerns | 52.61%             | 0.0949           | 0.0489           |
| Review Effectiveness of total Problems | 52.99%             | 0.0661           | 0.0502           |
| Average of Impact per Concern          | 11.14%             | 0.5957           | 0.0230           |
| Average of Impact per Problem          | 9.61%              | 0.6307           | 0.0430           |
| Review Cost Effectiveness of Concerns  | 75.04%             | 0.0060           | 0.0218           |
| Review Cost Effectiveness of Problems  | 75.01%             | 0.0048           | 0.0233           |

## 6. THREATS TO VALIDITY

The 22 projects in this experiment cover different applications with diverse technical characteristics and different clients. Also, reviewers with different personnel capabilities and non-uniform granularity of issues reported by different reviewers can also impact the number of issues reported and reviewing effectiveness displayed in this experiment. These certainly contributed to the large standard deviations seen for some of the results. Even so, the combination of both qualitative and quantitative evidence produced viable conclusions for some of the outcomes.

## 7. RELATED WORK

Up to date, many focused review or reading methods and techniques have been proposed, practiced and proved to be superior to unfocused reviews. The most common one in practice is checklist-based reviewing (CBR) [20], others include perspective-based reviewing (PBR) [21, 22], defect-based reading [23] (DBR), functionality-based reading (FBR) [24] and usage-based reading (UBR) [25, 26]. However, Most of them are value-neutral (except UBR) and focused on one single aspect, e.g. DBR focuses defect classification to find defects in artifacts and a scenario is a key factor in DBR. UBR focuses on prioritizing use cases in order of importance from a user perspective. FBR is proposed to trace framework requirements to produce well-constructed framework and review the code. Our method is a



more systematic value-based prioritization because it considers importance for the client and the whole project health/progress, fault-proneness for artifact developers and also reviewing costs for reviewers from different perspectives of stakeholders and few empirical studies have been done to prioritize artifacts to the best of our knowledge.

## 8. CONCLUSION

As the next generation systems/software process model, Incremental Commitment Spiral Model adopts value-based, risk driven concept as its basis, and this paper provides an empirical study on its application on V&V process in real-client projects. As an extension of the earlier Value-based V&V process development [6], the multi-criteria artifact-based review-prioritization process successfully enabled reviewers to prioritize artifact reviews and achieve strong improvement results. The qualitative and quantitative evidences observed from the two-year comparative experiment showed with statistical significance that the value-based artifact prioritization enabled reviewers to capture issues with high impact in a timely manner, and to improve the cost-effectiveness and ROI of reviewing activities.

## 9. ACKNOWLEDGEMENT

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**Table 16. An example of value-based artifact prioritization**

| <b>Weights:</b> | <b>1</b>                                                                                                                                                                                                                     | <b>1</b>                                                                                                                                                                                                                | <b>1</b>                                    | <b>1</b>                                                                                                       |                 |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|----------------------------------------------------------------------------------------------------------------|-----------------|
|                 | <b>Importance</b>                                                                                                                                                                                                            | <b>Quality Risk</b>                                                                                                                                                                                                     | <b>Dependency</b>                           | <b>Review Cost</b>                                                                                             | <b>Priority</b> |
| <b>LCP</b>      | <b>M</b><br>This document describes the life cycle plan of the project. This document serves as supporting function, without this, the project still could move on. With his document, the project could move more smoothly. | <b>L</b><br>Based on previous reviews, the author of this document has a strong sense of responsibility.                                                                                                                | <b>L</b>                                    | <b>M</b><br>A lot of new sections added, but this document is not very complex.                                | 1.00            |
| <b>OCD</b>      | <b>H</b><br>This document gives the overall operational concept of the system. This document is important, but it is not critical for this success of the system.                                                            | <b>VH</b><br>This is a complex document and a lot of the sections in this document needed to be redone based on the comments received from the TA.                                                                      | <b>M</b><br>SSRD                            | <b>H</b><br>Old document, but a lot of rework done.                                                            | 1.67            |
| <b>FED</b>      | <b>H</b><br>This document should be rated high because it provides feasibility evidence for the project. Without this document, we don't know whether the project is feasible.                                               | <b>H</b><br>The author of this document does not have appropriate time to complete this document with quality work.                                                                                                     | <b>H</b><br>SSRD, SSAD                      | <b>H</b><br>A lot of new section added to this version of the document.                                        | 1.00            |
| <b>SSRD</b>     | <b>VH</b><br>This document contains the requirements of the system. Without this document, the project can't move forward or even fail.                                                                                      | <b>VH</b><br>This is a complex document. This document needs to be consistent with win conditions negotiation, which might not be complete at this point. Also, a lot of rework was required based on comments from TA. | <b>L</b>                                    | <b>VH</b><br>This is an old document, but it is complex with a lot of rework.                                  | 2.50            |
| <b>SSAD</b>     | <b>VH</b><br>This document contains the architecture of the system. Without this document, the project can't move forward or even fail.                                                                                      | <b>VH</b><br>This is a complex document and it is a new document. The author of this document did not know that this document was due until the morning of the due date.                                                | <b>H</b><br>SSRD, OCD                       | <b>VH</b><br>This is an old document, but it is complex with a lot of rework done for this version.            | 1.25            |
| <b>SID</b>      | <b>VL</b><br>This document serves as supporting function, without this document, the project still could move on, but the project could move on more smoothly with this document.                                            | <b>L</b><br>This is an old document. Only additions made to existing sections.                                                                                                                                          | <b>VH</b><br>OCD, SSRD, FED, LCP, SSAD, PRO | <b>VL</b><br>This is an old document and this document has no technical contents.                              | 0.40            |
| <b>PRO</b>      | <b>H</b><br>Without this document, the project can probably move forward, but the system might not be what the customer is expecting. This document allows the customer to have a glimpse of the system.                     | <b>L</b><br>This is an old document with little new contents. The author has a high sense of responsibility and he fixed bugs from the last review in reasonable time.                                                  | <b>M</b><br>FED                             | <b>L</b><br>This is an old document with little content added since last version and not much rework required. | 1.33            |